

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 2nd Year Backlog Examination-2021
LE 2201
Leather Manufacturing Technology III

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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| 1(a) Define wet blue leather. Describe the procedure of boil test of wet blue leather. | 08 |
| 1(b) State the importance of rechroming process for shoe upper leather. | 07 |
| 1(c) Write down the objectives of deacidification in leather processing. | 05 |
| 1(d) Mention some neutralizing agents with their properties. | 09 |
| 1(e) Briefly discuss the effect of deacidification prior to dyeing and fat liquoring. | 06 |
| | |
| 2(a) Assume you have been given goat wet blue with shaved thickness 1.0 mm for shoe upper leather. Formulate a recipe to conduct deacidification, dyeing and fat liquoring. | 09 |
| 2(b) Define the terms: (i) Dye, (ii) Primaries and (iii) Intermediates. | 06 |
| 2(c) Distinguish between acid and basic dye with examples. | 08 |
| 2(d) State the role of leveling and penetrating agents in leather dyeing. | 06 |
| 2(e) Write down the objectives of the following mechanical operations: (i) Sammying, (ii) Splitting and (iii) Shaving. | 06 |
| | |
| 3(a) How do you select dye for specific leather production? Explain in detail. | 07 |
| 3(b) If you are provided a very loose wet blue leather- how will you select retanning agents for making shoe upper leather? Explain in detail. | 08 |
| 3(c) Explain dye fixing mechanism. | 04 |
| 3(d) Describe the parameters to be controlled during dyeing. | 08 |
| 3(e) As a leather engineer- state the selection criteria of fat liquoring for shoe upper, gloving and garments leather. | 08 |
| | |
| 4(a) Define color fastness of dye. Explain the significance of color fastness of dye in leather manufacturing. | 07 |
| 4(b) Concisely state 1:2 metal complex dye in leather processing. | 05 |
| 4(c) Propose a standard sole leather preparation recipe. | 06 |
| 4(d) Define the terms: (i) Gum, (ii) Tan and (iii) Non-tan. | 06 |
| 4(e) Discuss the common vegetable tanning agent sources. | 06 |
| 4(f) Vegetable tanned leather is less biodegradable than chrome tanned leather. Explain. | 05 |

SECTION-B

5(a)	Demonstrate the fixing mechanism of aldehyde, oil, resin and syntan in tabular form.	08
5(b)	Differentiate between Nerodol and Novolok type syntan.	06
5(c)	Briefly discuss on some factors that may affect the tanning power of syntan.	10
5(d)	Define the following terms: (i) Retanning, (ii) Crust leather and (iii) Chamois leather.	06
5(e)	Write down the properties of resin tanning agent.	05
6(a)	Write down the influence of oxazolidine on chrome tanning.	05
6(b)	Differentiate between formaldehyde and glutaraldehyde tanning agent.	06
6(c)	Briefly, state about the aldehyde-protein interaction.	10
6(d)	Define oil tanning. Why aldehyde tanning is done before oil tanning?	10
6(e)	What are the health and safety implications associated with formaldehyde tanning?	08
7(a)	Write on the principle of oil tanning.	05
7(b)	Briefly discuss on the selection criteria of fish oil for the preparation of chamois leather.	12
7(c)	State the mechanism of syntan interaction with collagen.	06
7(d)	Briefly discuss the aldehyde and adsorption theory of oil tanning.	08
7(e)	State the application of chamois leather.	04
8(a)	Define color safe bleaching method of chamois leather. Why oxidative bleaching is not suitable for bleaching chrome/vegetable tanned leather?	07
8(b)	Describe the bleaching method of chamois leather.	05
8(c)	State the key objectives of the following operations: (i) Vibrating staking, (ii) Vacuum drying, (iii) Toggle drying and (iv) Overhead drying.	12
8(d)	Why conditioning is needed before vibrating staking?	06
8(e)	State the principle of bleaching.	05

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Leather Manufacturing Technology II

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Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) Define liming. Write down the objectives of liming. 08
- 1(b) Classify unhairing. Briefly discuss the oxidative method of unhairing with advantages and disadvantages. 14
- 1(c) State unhairing method of painting. If you are provided a new sheep skin-formulate a recipe to conduct soaking and painting. 08
- 1(d) Write down the behavior of collagen in acidic and alkaline media. 05
- 2(a) Define green fleshing. Write down the objectives of limed fleshing. 07
- 2(b) What is meant by iso-electric point? Discuss briefly the shift of iso-electric point of collagen due to liming. 10
- 2(c) Write down the composition of fleshing. State the environmental effect of fleshing. 08
- 2(d) Define sharpening agent. Mention some common sharpening agents. State the chemistry of unhairing. 10
- 3(a) Write short notes on (i) Lime blast and (ii) Immunization of keratin. 08
- 3(b) State the principle of hair saving and hair burning unhairing. 10
- 3(c) Write short notes on (i) Swelling and plumping and (ii) sweating. 08
- 3(d) List out some possible causes, effects, and remedies of some common liming defects. 09
- 4(a) Define deliming. Write down the objectives of deliming. 06
- 4(b) State carbon dioxide deliming with advantages and limitations. 10
- 4(c) Briefly discuss the chemistry of deliming with associated chemical reaction of commercial deliming agents. 10
- 4(d) Briefly discuss environmental effect of limed fleshing. 09

SECTION-B

- 5(a) What are enzymes? Write on some common enzymes used in the commercial bating preparation. 08
- 5(b) State the effect of temperature, pH and origin of enzymes on the bating operation. 09
- 5(c) Masking agent is necessary during chrome tanning- Explain. 06
- 5(d) Write on the mechanism of the following degreasing methods: (i) Aqueous emulsification and (ii) Enzymatic degreasing. 12

6(a)	Define pickling. Write down the role of pickling in leather manufacturing.	07
6(b)	Briefly discuss the various parameters responsible for the process control of the pickling.	12
6(c)	What is acid swelling? How can it be minimized during pickling operation?	08
6(d)	Define non swelling acids with examples. State the functions of non-swelling acids in pickling process.	08
7(a)	What is tanning? Write down the features of tanning.	07
7(b)	Write on the ionic variations of the chrome tanning salt.	09
7(c)	Calculate the proctor, Schorlemmer and Frieberg basicity of the chrome complex having the formula of $\text{Cr}_8(\text{SO}_4)_6(\text{OH})_{12}$.	08
7(d)	Briefly discuss on the factors influencing the chrome tanning process.	11
8(a)	State the reaction mechanism of chromium salt with skin collagen.	07
8(b)	Write notes on the self-basifying chrome tanning agent.	08
8(c)	Write about the effect of masking in chrome tanning.	10
8(d)	Differentiate between single bath and double bath chrome tanning process.	10